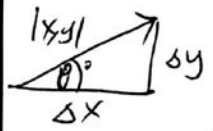
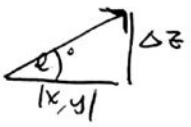
Azimuth

X-y plane

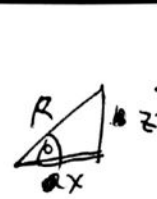


elevation



longitude [°]

xyz plane



$\sin(\varphi) = z/R$

$\cos(\chi) = \cancel{\quad}$

$\sin(\varphi) = z/R$

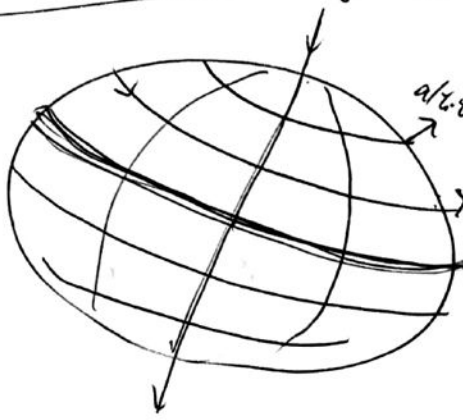
$\cos(\varphi) = x/R$ (φ on y/R)

X-y plane

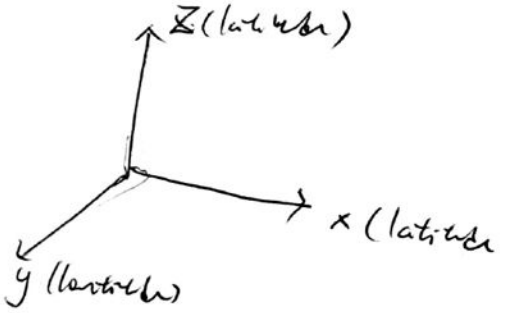


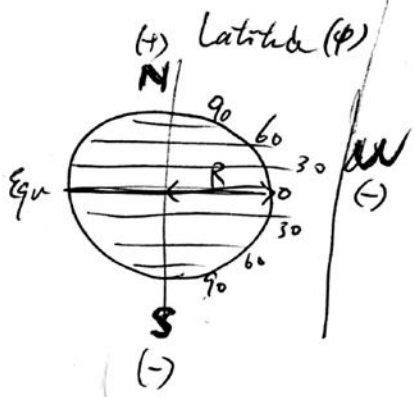
$\sin(\theta\lambda) = y/R$

$\cos(\theta\lambda) = x/R$



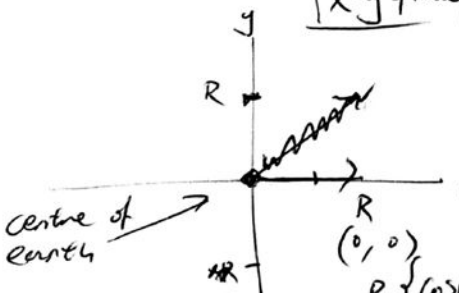




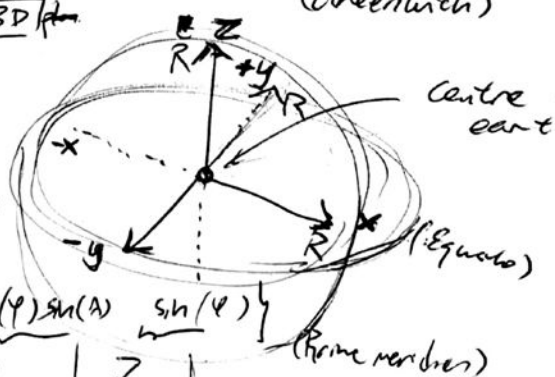




X-y plane



3D



$R \{ \cos(\varphi)\cos(\lambda) \quad \cos(\varphi)\sin(\lambda) \quad \sin(\varphi) \}$

φ	λ	R	x	y	z
0°	0°	R	R	0	0
0°	90°		0	R	0
90°	any		0	0	R
-90°	any		0	0	-R
45°	45°		R/2	R/2	R/√2

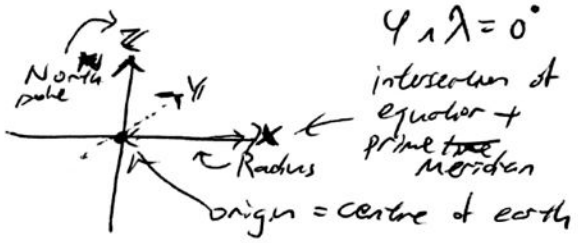
ensures alignment w/ earth's curvature

ECEF (Eearth centred earth fixed) Cartesian coordinates

Earth Centred Earth Fixed $\xrightarrow{\{Tr\}}$ East North Up

↳ translate the sat's position relative to the observer into a local reference frame.

① ECEF is a global Cart. sys.

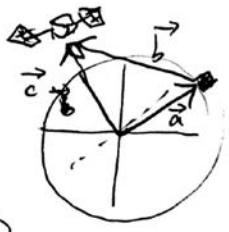


$\varphi, \lambda = 0°$ intersection of equator + prime meridian

$$\begin{cases} x = (R+h)\cdot\cos\varphi\cdot\cos\lambda \\ y = (R+h)\cdot\cos\varphi\cdot\sin\lambda \\ z = (R+h)\cdot\sin\varphi \end{cases}$$

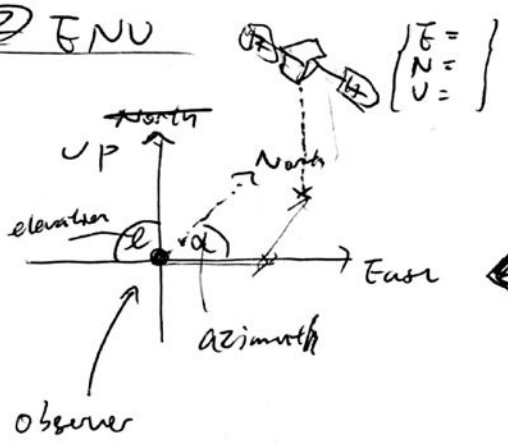
φ: latitude°, λ: longitude°

R: radius of earth, h: altitude (km)



$$\vec{b} = \vec{c} - \vec{a} = \begin{pmatrix} x_c - x_a \\ y_c - y_a \\ z_c - z_a \end{pmatrix} = \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \end{pmatrix}$$

② ENU



$$e = \arctan\left(\frac{U}{\sqrt{E^2+N^2}}\right)$$

$$\alpha = \arctan\left(\frac{E}{N}\right)$$

Rotation matrix to align the $\vec{b}$ in ENU domain.

$$\begin{bmatrix} E \\ N \\ U \end{bmatrix} = \begin{bmatrix} -\sin(\lambda) & \cos(\lambda) & 0 \\ -\sin(\varphi)\cos(\lambda) & -\sin(\varphi)\sin(\lambda) & \cos(\varphi) \\ \cos(\varphi)\cos(\lambda) & \cos(\varphi)\sin(\lambda) & \sin(\varphi) \end{bmatrix} \cdot \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

φ: latitude of observer, λ: longitude of observer.

ECEF → ENU ?

$$\vec{b} = \vec{c} - \vec{a}$$

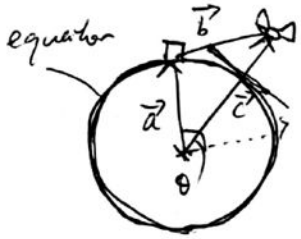
$$\vec{b} = \begin{Bmatrix} x \\ y \\ z \end{Bmatrix}$$

dot product

$$\cos\theta = \frac{a \cdot b}{|a||b|}$$

$$\cos e = \frac{a \cdot b}{|a||b|}$$

angle of elevation

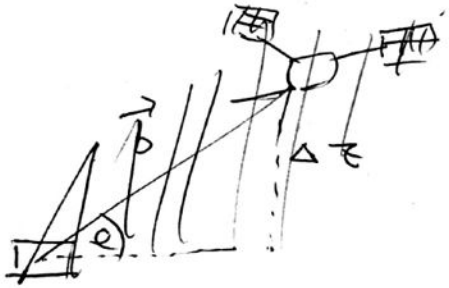


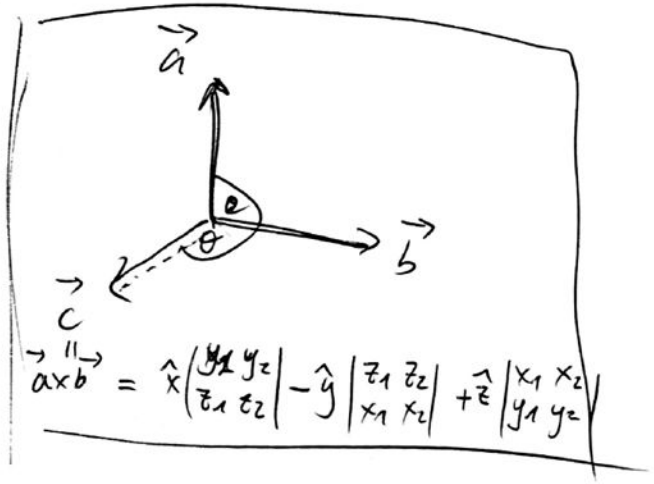
~~bigger motor / south~~
~~rotate~~
~~lower motor~~ X

a) assume xy plane
→ φ (latitude) = 0°

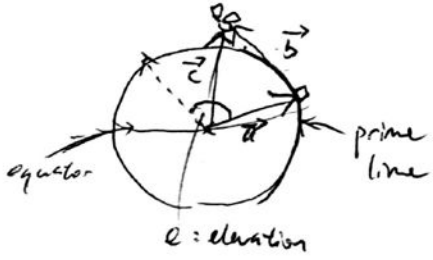
(azimuth)

* θ is the angle of horizontal rotation for the bottom motor





$$\vec{a} \times \vec{b} = \hat{x}\begin{vmatrix} y_1 & y_2 \\ z_1 & z_2 \end{vmatrix} - \hat{y}\begin{vmatrix} z_1 & z_2 \\ x_1 & x_2 \end{vmatrix} + \hat{z}\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}$$



e : elevation





